

IoT-Based PM2.5 And PM10 AIR QUALITY MONITORING SYSTEM

SOMNATH BERA

This is an ESP32 NodeMCU-based environmental monitoring system to check particulate matter (PM), humidity, temperature, altitude and pressure levels. Various environmental conditions of the place are tested and displayed on the TFT display as well as on ThingSpeak IoT (Internet of Things) platform. The author's prototype is shown in Fig. 1.

Particulate matter

High concentrations of dust or PM is a serious health concern. PM2.5 is less than 2.5 microns in diameter, and PM10 is less than 10 microns

in diameter. This means, a PM10 report includes PM2.5 as well. Both these particles are much smaller than a human hair, which is about 70 microns in width.

PM10. Operations such as stone crushing, coal grinding, rotary kilning in cement industry and dust on road stirred by moving vehicles can increase PM10 levels. PM10 limit for 24-hour average is $150\mu\text{g}/\text{m}^3$.

PM2.5. This is a result of fine particles produced from all types of combustion, including motor vehi-

cles, thermal power plants, residential wood burning, forest fires, agricultural burning and other industrial processes. PM2.5 limit for 24-hour average is $35\mu\text{g}/\text{m}^3$.

Circuit and working

The circuit diagram of the IoT-based air quality monitoring system is shown in Fig. 2. It consists of ESP32 NodeMCU, SDS011, BME280, ILI9163 TFT display, a resistor and transistor 2N2219. ESP32 NodeMCU is the heart of the circuit. Connection details of various modules to ESP32 board are given in tables I, II and III.

ESP32 NodeMCU. It is a low-power microcontroller (MCU) with integrated Wi-Fi and dual-mode Bluetooth—it is a successor of ESP8266 MCU. ESP32 is shown in Fig. 3. Arduino IDE is used to upload the sketch to ESP32 NodeMCU. (ESP-Wroom-32 version 1.1 was used at EFY Lab for testing purpose.)

TFT display. A 3.65cm

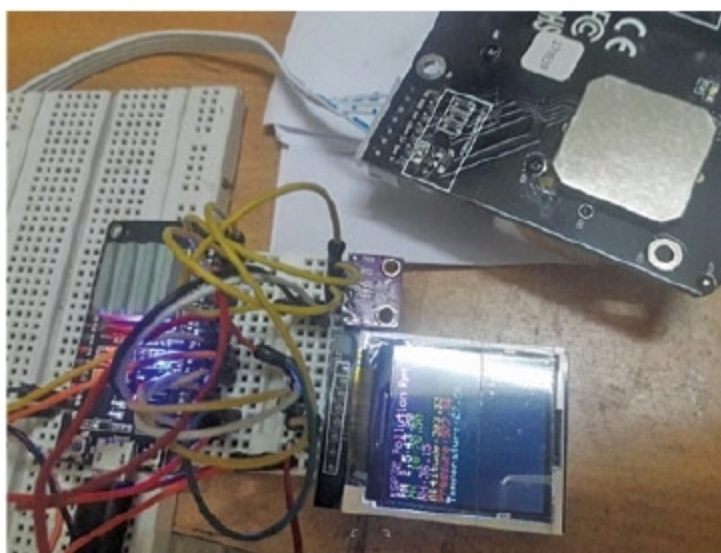


Fig. 1: Author's prototype

TABLE I: CONNECTIONS BETWEEN ILI9163 AND ESP32	
ILI9163	ESP32
LED, V_{CC}	3.3V
GND	GND
A0	GPIO15
CS	GPIO2
RST	GPIO4
SCK	GPIO18
SDA	GPIO23

TABLE II: CONNECTIONS BETWEEN SDS011 AND ESP32	
SDS011	ESP32
5V	5V
GND	GND
TX	GPIO16 (P16)
RX	GPIO17 (P17)

TABLE III: CONNECTIONS BETWEEN BME280 AND ESP32	
BME280	ESP32
V_{CC}	3.3V
GND, SDO	GND
SCL	GPIO22
SDA	GPIO21
CSB	NC

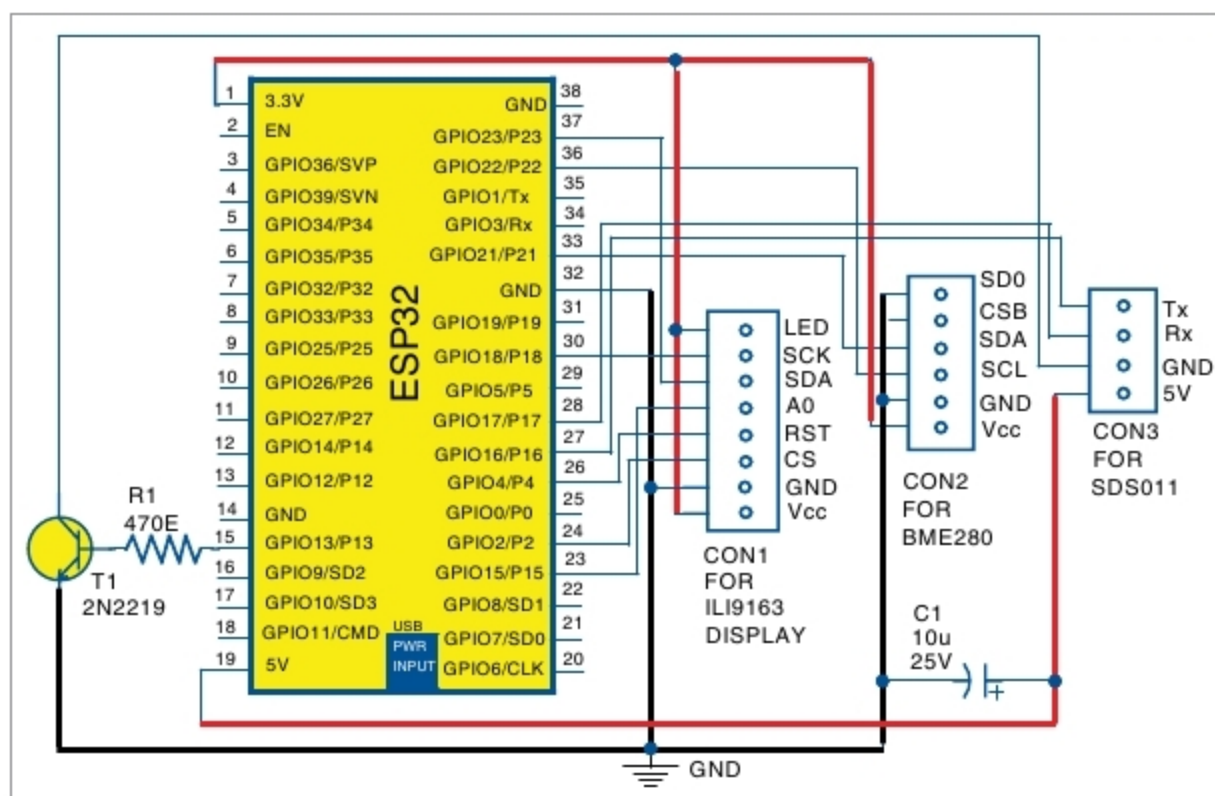


Fig. 2: Circuit diagram of ESP32-based air quality monitoring system